Karthik Abinav Sankararaman

October 2026

Contact

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Interests

LLMs & AI Foundations, Applications and Systems: Sequential Decision Making, Sequential Deep Learning, Optimization, Matching-based Market Design (Rideshare, Online Advertising, Modern Rec Sys), AI for Society (Fairness, Integrity, Climate Change), Large Scale ML Systems, Generative AI.

Education

University of Maryland, College Park
PhD. in Computer Science — **September 2014 - July 2019**
M.S. in Computer Science — **December 2016**

Indian Institute of Technology, Madras — **August 2010 - July 2014**
B.Tech Honours in Computer Science and Engineering
Minor: Operations Research

Awards

- *Top-5 most downloaded papers from TEAC '21 & '22*
- *Best/Top/Outstanding reviewer*, NeurIPS 2019, ICML 2020, NeurIPS 2021
- *Future Faculty Fellow*, University of Maryland, 2018
- *Dean's Fellowship*, University of Maryland, 2014, 2015
- *S.N. Bose Scholarship*, India, 2013
- *National Talent Search (NTSE)* Scholarship, India, 2010

Recent Work and Research Experience

Facebook/Meta, Menlo Park, USA — **September 2019 -**
AI Research in Meta Superintelligence Labs
RL, data flywheel, model behavior & EQ, Agents. Previously support teams to develop new algorithms and systems based on RL, LLMs, bandits, active learning across many problem teams (ads, recommender systems, ML dev tools, Safety) within the company.

Microsoft Research New York City, USA — **June - September 2018**
Supported by Nicole Immorlica, Rob Schapire, Alex Slivkins

Indian Institute of Science & Microsoft Research, Bengaluru, India — **May - July 2017**
Supported by Anand Louis, Navin Goyal

IBM Almaden Research Center, San Jose, USA — **May - August 2016**

Adobe Inc., San Jose, USA — **May - August 2015**

Research Assistant at UMD, College Park, USA — **August 2014 - July 2019**

Teaching Experience

Instructor, University of Maryland — **January - May 2019**
CMSC250H - Discrete Structures (Honors) along with Prof. Bill Gasarch
Responsibilities: Designing the syllabus, homework, exams and grading policy. Weekly lecturing.

Teaching Assistant, University of Maryland — **August 2014 - December 2018**
CMSC250 - Discrete Structures (2 sems.), CMSC131- Intro to Programming (2 sems.), CMSC451/651- Advanced Algorithms (5 sems.)
Responsibilities: Guest Lectures, Conducting Discussion Sessions, Office Hours, Grading

Teaching Assistant, Indian Institute of Technology, Madras — **January - April 2014**
Paradigms of Programming
Responsibilities: Grading Programming Assignments

Programming

C++, Python, SQL/Presto/Hive, PyTorch

PUBLICATIONS (HIGHLY PEER-REVIEWED) J: JOURNAL, C: CONFERENCE

1. "Generalized Parallel Scaling with Interdependent Generations" — Joint work with Harry Dong, David Brandfonbrener, Eryk Helenowski, Yun He, Mrinal Kumar, Han Fang, Yuejie Chi
   - *(C) The 14th International Conference on Learning Representations (ICLR), 2026*
   - *(W) Best Paper Award nomination at The First Workshop on Efficient Reasoning, NeurIPS 2025*
2. "Learning Auxiliary Tasks Improves Reference-Free Hallucination Detection in Open-Domain Long-Form Generation" — Joint work with Chengwei Qin, Wenxuan Zhou, Nanshu Wang, Tengyu Xu, Alexander Radovic, Eryk Helenowski, Arya Talebzadeh, Aditya Tayade, Sinong Wang, Shafiq Joty, Han Fang, Hao Ma
   - *(C) The 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025) [Main Conference]*
3. "Think Smarter not Harder: Adaptive Reasoning with Inference Aware Optimization " — Joint work with Zishun Yu, Tengyu Xu, Di Jin, Yun He, Wenxuan Zhou, Zhouhao Zeng, Eryk Helenowski, Chen Zhu, Sinong Wang, Hao Ma, Han Fang
   - *(C) The 42nd International Conference on Machine Learning (ICML 2025)*
4. "Promoting External and Internal Equities Under Ex-Ante/Ex-Post Metrics in Online Resource Allocation" — Joint work with Pan Xu, Aravind Srinivasan
   - *(C) The 41st International Conference on Machine Learning (ICML 2024) [Spotlight - top 3.5% of accepted papers]*
5. "Effective Long-Context Scaling of Foundation Models" — Joint work with 21 authors
   - *(C) Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2024*
6. "Rethinking Incentives in Recommender Systems: Are Monotone Rewards Always Beneficial?" — Joint work with Fan Yao, Chuanhao Li, Yiming Liao, Yan Zhu, Qifan Wang, Hongning Wang, Haifeng Xu
   - *(C) The 37th Conference on Neural Information Processing Systems (NeurIPS), 2023*
7. "Contextual Bandits with Packing and Covering Constraints: A Modular Lagrangian Approach via Regression" — Joint work with Alex Slivkins, Dylan Foster, Xingyu Zhou
   - *(C) The 36th Annual Conference on Learning Theory (COLT), 2023*
   - *(J) Journal of Machine Learning Research (JMLR), 2024 (IF: 4.091)*
8. "Allocation Problem in Remote Teleoperation: Online Matching with Offline Reusable Resources and Delayed Assignments" — Joint work with Osnat Ackerman Viden, Yohai Trabelsi, Pan Xu, Oleg Maksimov and Sarit Kraus
   - *(C) The 22nd International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2023*
9. "Robust Identifiability in Linear Structural Equation Models for Causal Inference" — Joint work with Navin Goyal, Anand Louis
   - *(C) The 38th Conference on Uncertainty in Artificial Intelligence (UAI), 2022*
   - *NeurIPS Workshop on Safety and Robustness in Decision Making, 2019*
10. "Online minimum matching with uniform metric and random arrivals" — Joint work with Sharmila Duppala and Pan Xu
    - *(J) Operations Research Letters, 2022 (IF: 0.76)*
11. "Bandits with Knapsacks beyond the Worst-Case" — Joint work with Alex Slivkins
    - *(C) The 35th Conference on Neural Information Processing (NeurIPS), 2021*
12. "Beyond $\log^2(T)$ Regret for Decentralized Bandits in Matching Markets" — Joint work with Soumya Basu, Abishek Sankararaman
    - *(C) The 38th International Conference on Machine Learning (ICML), 2021*
13. "Multi-armed Bandits with Cost Subsidy" — Joint work with Deeksha Sinha, Abbas Kazerouni and Vashist Avadhanula
    - *(C) The 24th International Conference on Artificial Intelligence and Statistics (AIStats), 2021*
14. "Dominate or Delete: Decentralized Competing Bandits with Uniform Valuation" — Joint work with Abishek Sankararaman and Soumya Basu
    - *(C) The 24th International Conference on Artificial Intelligence and Statistics (AIStats), 2021*
15. "Stochastic bandits for multi-platform budget optimization in online advertising" — Joint work with Vashist Avadhanula, Riccardo Colini-Baldeschi, Stefano Leonardi and Okke Schrijvers
    - *(C) The 30th Web Conference 2021 (formerly known as WWW)*

16. "Analyzing the effect of neural network architecture on training performance" — Joint work with Soham De*, Zheng Xu, Ronny Huang, Tom Goldstein
    - *(C) The 37th International Conference on Machine Learning (ICML), 2020*
    - *NeurIPS Workshop on Integration of Deep Learning Theory, 2018*
17. "Matching Algorithms for Blood Donation" — Joint work with Duncan McElfresh, Sergey Pupyrev, Christian Kroer, John Dickerson, Eric Sodomka, Zack Chauvin and Neil Dexter
    - *(C) The 21st ACM Conference on Economics and Computation (EC) 2020*
    - *(J) Nature Machine Intelligence, 2023 (IF: 25.9)*
18. "Balancing the Tradeoff between Profit and Fairness in Rideshare Platforms during High-Demand Hours" — Joint work with Vedant Nanda, John Dickerson, Aravind Srinivasan, Pan Xu
    - *(C) The 34th AAAI Conference on Artificial Intelligence (AAAI), 2020*
    - *(C) Extended Abstract in the Third AAAI/ACM Conference on AI, Ethics and Society (AIES), 2020 (**Oral Presentation**)*
19. "Mix and Match: Markov Chains and Mixing Times for Matching in Rideshare" — Joint work with Mike Curry, John Dickerson, Aravind Srinivasan, Yuhao Wan, Pan Xu
    - *(C) The 15th Conference on Web and Internet Economics (WINE), 2019*
20. "Adversarial Bandits with Knapsacks" — Joint work with Nicole Immorlica, Robert Schapire, Alex Slivkins
    - *(C) The 60th IEEE Symposium on Foundations of Computer Science (FOCS), 2019*
    - *INFORMS workshop on Market Design (with EC 2019)*
    - *(J) Journal of the ACM (JACM), 2022 (IF: 2.35)*
21. "Stability of Linear Structural Equation Model of Causal Inference" — Joint work with Navin Goyal, Anand Louis
    - *(C) The 35th Conference on Uncertainty in Artificial Intelligence (UAI), 2019*
    - *NeurIPS Workshop on Causality, 2018*
22. "Online Resource Allocation with Matching Constraints" — Joint work with John Dickerson, Kanthi Sarpatwar, Aravind Srinivasan, Kun-Lung Wu, Pan Xu
    - *(C) The 18th Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2019*
23. "A Unified Approach to Online Matching with Conflict-Aware Constraints" — Joint work with Hao Cheng, John Dickerson, Yexuan Shi, Aravind Srinivasan, Yongxin Tong, Leonidas Tsepenekas, Pan Xu
    - *(C) The 33rd AAAI Conference on Artificial Intelligence (AAAI), 2019*
24. "Balancing Relevance and Diversity in Online Matching via Submodularity" — Joint work with John Dickerson, Aravind Srinivasan, Pan Xu
    - *(C) The 33rd AAAI Conference on Artificial Intelligence (AAAI), 2019*
25. "Assigning Workers to Tasks in Crowdsourcing Platforms: Two-Sided Online Matching" — Joint work with John Dickerson, Aravind Srinivasan, Pan Xu
    - *(C) The 17th Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2018*
    - *(J) Full version in Transactions on Economics and Computation (TEAC), 2024 (IF: 4.32)*

26. "Combinatorial Semi-Bandits with Knapsacks" — Joint work with Alexandrs Slivkins
    - *(C) The 21st International Conference on Artificial Intelligence and Statistics (AIStats), 2018 — **(Invited for Oral Presentation)***
27. "Allocation Problems in Ride-Sharing Platforms: Online Matching with Offline Reusable Resources" — Joint work with John Dickerson, Aravind Srinivasan, Pan Xu
    - *(C) The 32th AAAI Conference on Artificial Intelligence (AAAI), 2018 — **(Invited for Oral Presentation)***
    (J) Full version in Transactions on Economics and Computation (TEAC), 2021 (IF: 4.32)
    (Award): Top-5 most downloaded papers, among papers published in TEAC '21 & '22 and Top-20 most downloaded among all papers published in TEAC
28. "Algorithms to Approximate Column-Sparse Packing Problems" — Joint work with Brian Brubach, Aravind Srinivasan, Pan Xu
    - *(C) The 29th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), 2018*
    - *(J) Full version in ACM Transactions of Algorithms (TALG), 2019 (IF: 0.4)*
29. "Attenuation-based Frameworks for Online Stochastic Matching with Timeouts" — Joint work with Brian Brubach, Aravind Srinivasan, Pan Xu
    - *(C) The 16th Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2017*
    - *(J) Full version in Algorithmica, 2019 (IF: 0.49)*

30. "Improved Algorithms for Online Stochastic Matching"— Joint work with Brian Brubach, Aravind Srinivasan, Pan Xu
    - *(C) The 24th Annual European Symposium on Algorithms (ESA), 2016*
    - *(J) Full version in Algorithmica 2020 (IF: 0.49)*
31. "Ensuring Privacy in Location-Based Services: An Approach Based on Opacity Enforcement" — Joint work with Yi-Chin Wu, Stèphane Lafortune
    - *(C) The 14th International Workshop of Discrete Event Systems (WODES), 2014*
32. " On the Equivalence of Graph Convolution and Mixup" — Joint work Xiaotian Han, Hanqing Zeng, Yu Chen, Shaoliang Nie, Jingzhou Liu, Kanika Narang, Zahra Shakeri, Song Jiang, Madian Khabsa, Qifan Wang, Xia Hu
    - *(J) Transactions of Machine Learning Research (TMLR), 2024*
    - *(C) Learning on Graphs, 2024* ***(Oral Presentation)***

THESIS

1. "Sequential Decision Making with Limited Resources"
   PhD Thesis, University of Maryland College Park, 2019

MANUSCRIPTS

1. "Self-Credit: Credit Assignment for Long-Horizon Agentic RL via Intra-group Rescaling" — Joint work with Fangcong Yin, Ming Yin, Tanvi Bhandarkar, Aashu Singh, Eryk Helenowski, Shlok Kumar Mishra, Saghar Hosseini, Greg Durrett, Dinkar Juyal
   - *Arxiv 2026*
2. "Muse Glimmer 30B: An On-Device Agentic Model" — Joint work with the Muse Glimmer team
   - *Arxiv 2026*
3. "Dual-Weighted Reinforcement Learning for Generative Preference Modeling" — Joint work with Shengyu Feng, Yun He, Shuang Ma, Beibin Li, Yuanhao Xiong, Songlin Li, Karishma Mandyam, Julian Katz-Samuels, Shengjie Bi, Licheng Yu, Hejia Zhang, Han Fang, Riham Mansour, Yiming Yang, Manaal Faruqui - *Arxiv 2025*
4. "Reinforcement Learning from User Feedback" — Joint work with Eric Han, Jun Chen, Xiaoliang Peng, Tengyu Xu, Eryk Helenowski, Kaiyan Peng, Mrinal Kumar, Sinong Wang, Han Fang, Arya Talebzadeh - *Arxiv 2025*
5. "Multi-IF: Benchmarking LLMs on Multi-Turn and Multilingual Instructions Following" — Joint work with Yun He, Di Jin, Chaoqi Wang, Chloe Bi, Karishma Mandyam, Hejia Zhang, Chen Zhu, Ning Li, Tengyu Xu, Hongjiang Lv, Shruti Bhosale, Chenguang Zhu, Eryk Helenowski, Melanie Kambadur, Aditya Tayade, Hao Ma, Han Fang, Sinong Wang - *Arxiv 2024*
6. "Preference Optimization with Multi-Sample Comparisons" — Joint work with Chaoqi Wang, Zhuokai Zhao, Chen Zhu, Michal Valko, Xuefei Cao, Zhaorun Chen, Madian Khabsa, Yuxin Chen, Hao Ma, Sinong Wang - *Arxiv 2024*
7. "The Perfect Blend: Redefining RLHF with Mixture of Judges" — Joint work with Tengyu Xu, Eryk Helenowski, Di Jin, Kaiyan Peng, Eric Han, Shaoliang Nie, Chen Zhu, Hejia Zhang, Wenxuan Zhou, Zhouhao Zeng, Yun He, Karishma Mandyam, Arya Talabzadeh, Madian Khabsa, Gabriel Cohen, Yuandong Tian, Hao Ma, Sinong Wang, Han Fang - *Arxiv 2024*
8. "BayesFormer: Transformer with Uncertainty Estimation" — Joint work with Sinong Wang and Han Fang - *Arxiv 2022*
9. "Improved Adaptive Algorithm for Scalable Active Learning with Weak Labeler" — Joint work with Yifang Chen, Alessandro Lazaric, Matteo Pirotta, Dmytro Karamshuk, Qifan Wang, Karishma Mandyam, Sinong Wang, Han Fang - *Arxiv 2022*
10. "Improved Approximation Algorithms for Stochastic-Matching Problems" — Joint work with Marek Adamczyk, Brian Brubach, Fabrizio Grandoni, Aravind Srinivasan and Pan Xu - *Arxiv 2021*

Voluntary Service

Senior Program Committee/Area Chair/Meta-reviewer. IJCAI '21 - '24, AAMAS '22, ACML '22 - '23, AAAI Safe and Robust AI Track '23, AAMAS '25, '26

Conference (reviewer and program committee). EC, NeurIPS, ICLR, AIStats, ICML, UAI, SODA, AAAI, WWW, IJCAI, COLT, CIKM, STOC, AI for Social Impact @ AAAI, AI for social good @ NeurIPS, NeurIPS Reproducibility Challenge, Global Challenges in EC

Journal (reviewer). Transactions on Algorithms, Networks, Optimization Letters, JAAMAS, Mathematics of Operations Research, Operations Research, Transactions of Signal Processing, Discrete Optimization, Artificial Intelligence, Nature-Scientific Reports, IEEE Transactions on Neural Networks and Learning Systems, Journal of Machine Learning Research (JMLR), Journal of Artificial Intelligence (JAIR), Journal of the ACM (JACM), Neural Computation

Mentor in "New in ML" workshop: Mentoring new researchers on writing machine learning papers.

Session Chair AAMAS '22